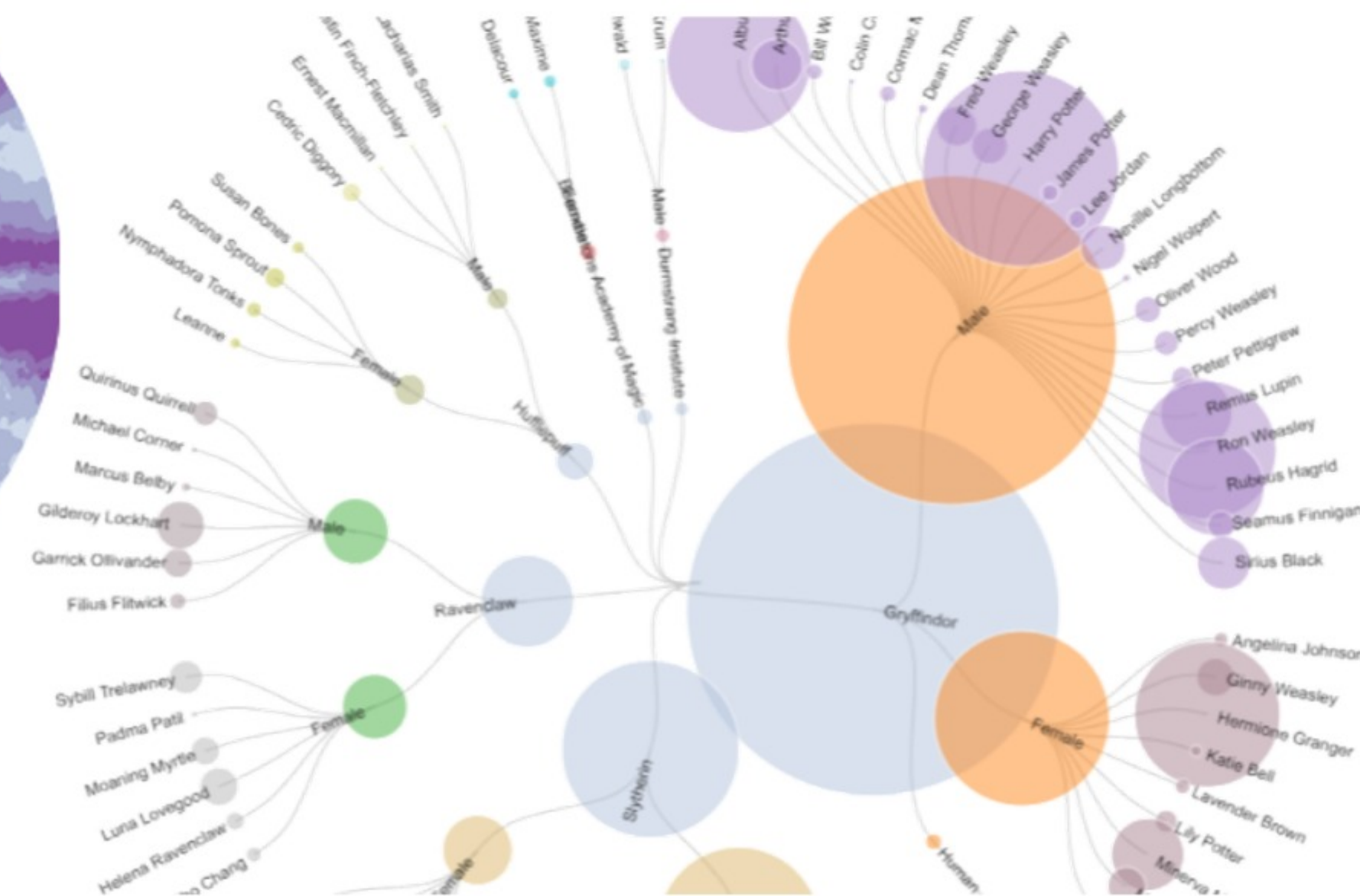
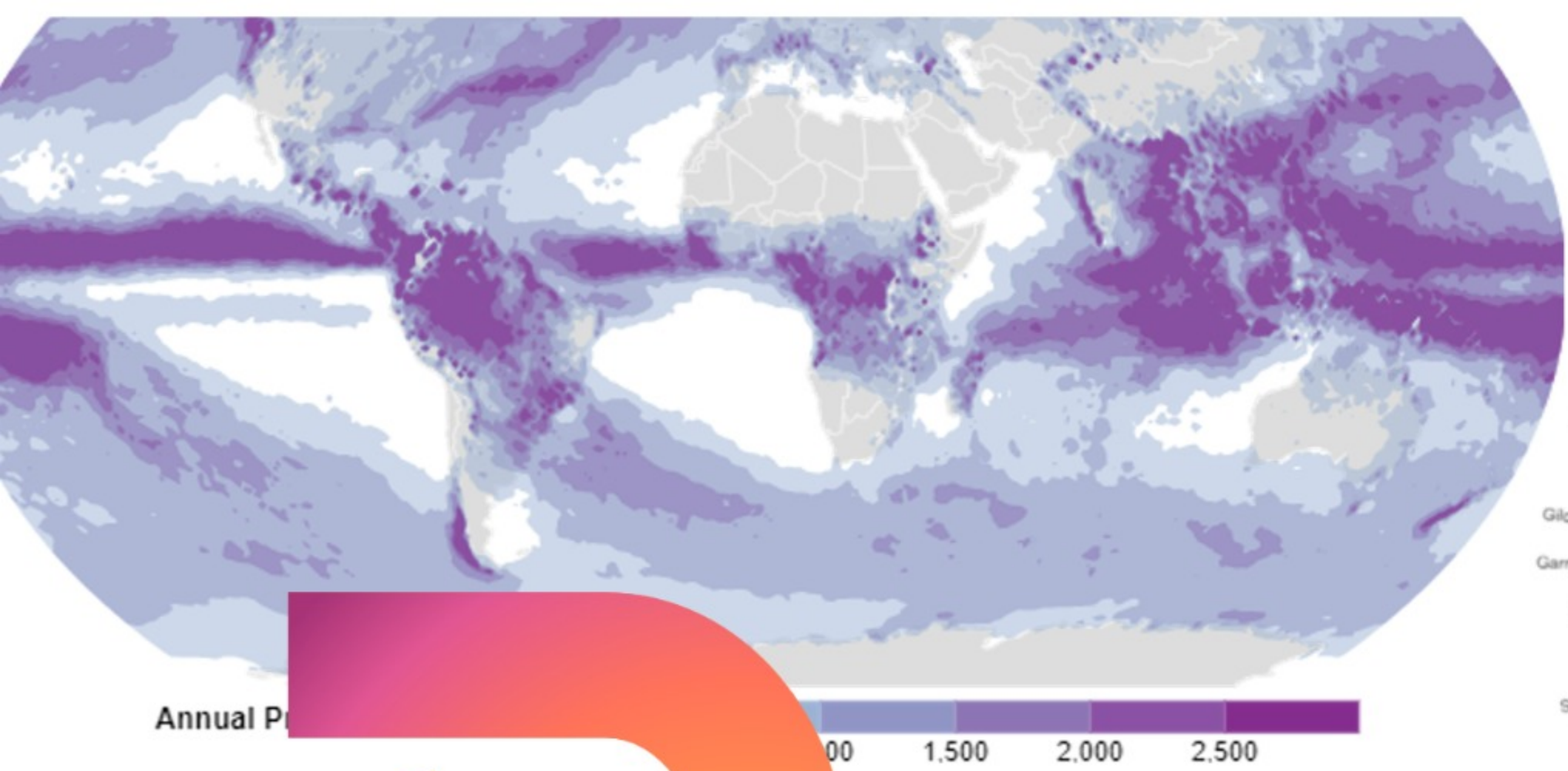




DENEB

Deneb in a Day

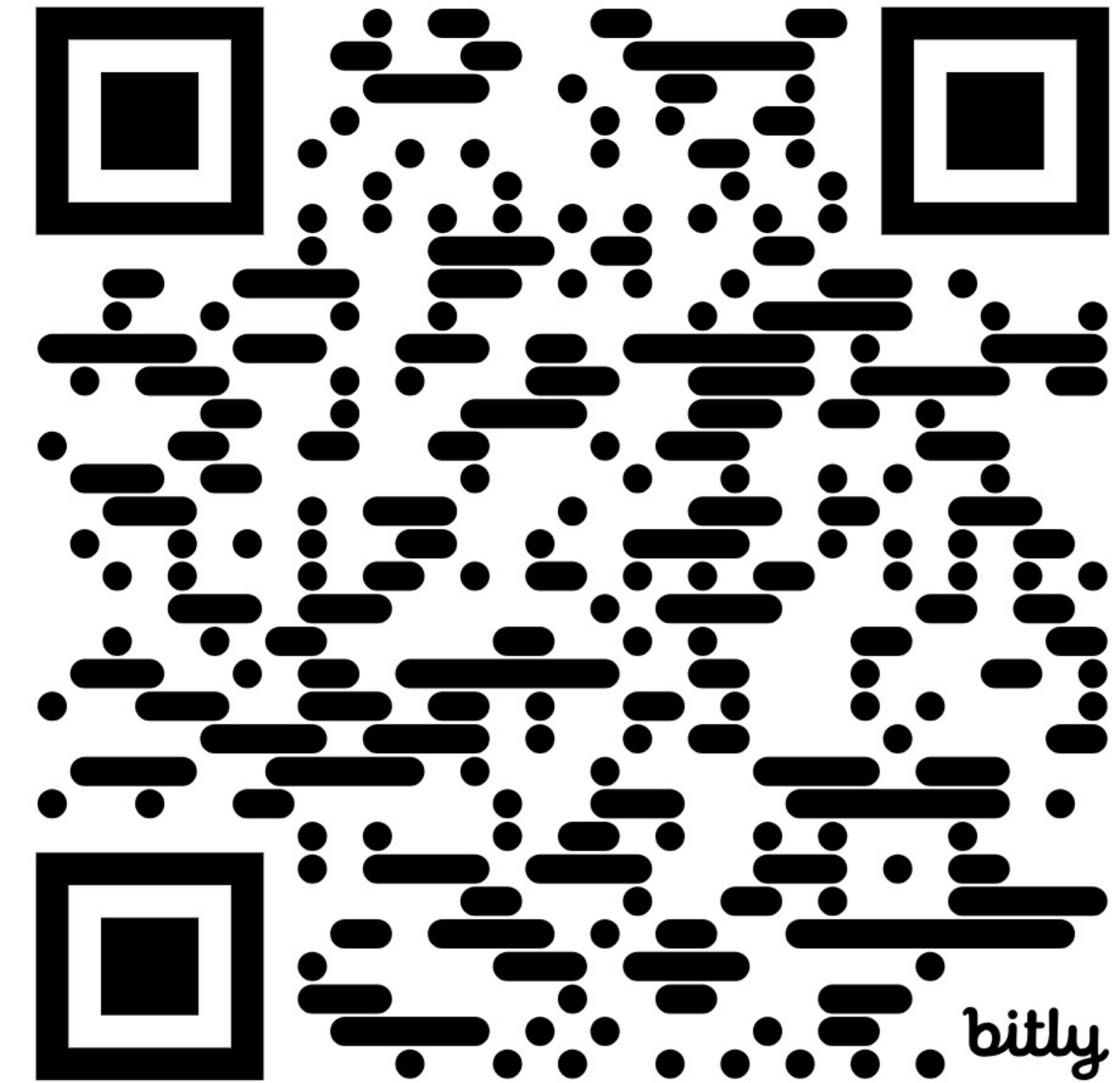
Grammer Language for Visuals

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Downloads

Resources

- Power BI (PBIX) sample report
- Presentation Slides
- Exercise Worksheets



Download at <https://bit.ly/deneb-02>

D

Hello!



Training.tips

Solita Germany

Welcome to



ben ferry



Solita Germany



Training.tips

Speakers

ben ferry



mike carlo



 [/powerbiferrytales](#)

 [@PowerBltips](#)

 [/michaelcarlo](#)

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Why are you here?

Goals

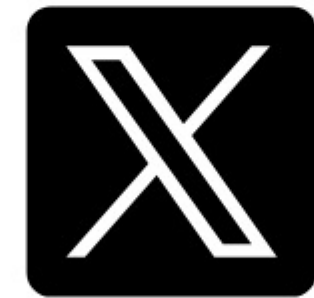
- What is Vega and Vega-Lite
- Build Visuals
- Understand where to find examples
- Hands on Experience
- Entertainment

Agenda

- Visual Grammer Language
- Overview of Vega-Lite
- Touch on common terms
- Demos & Builds
- Eat
- Food Coma
- Build & Demos
- Closing

Daniel Marsh-Patrick

the creator



@the_d_mp



/daniel-m-p



Deneb: Declarative
Visualization in Power BI



HTML Content (lite)



HTML Content



Violin Plot



Small Multiple Line Chart



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Introduction to



Lite
Vega

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Vega A Visualization Grammar



<https://vega.github.io/vega/>



<https://vega.github.io/vega-lite/>

<https://vega.github.io/vega/about/>

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JSON is your Friend

Data Structure

```
1 // JSON Comment
2 {
3     "name": "Mike",
4     "age": 30,
5     "colors": ["red", "blue", "green"]
6 }
```

Curly Brackets = { }

Bracket = []

<https://www.json.org/json-en.html>

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JSON - Objects and Arrays

Objects: unordered
collections in { }

Arrays: ordered lists in []

Can be combined for
complex data

```
1 // Object Example
2 {
3     "name": "Mike",
4     "age": 30
5 }
6
7 // Array Example
8 ["red", "blue", "green"]
```

JSON - Key Value Pairs

Element

```
1  {  
2    "name": "mike",  
3    "age": 30,  
4    "verified": true,  
5    "middle name": null,  
6    "scores": [ 95 , 100 ]  
7  }
```

Keys are strings in double quotes

Values can be string, number, boolean, null, object, or array

Value

```
1  // Object Example  
2  {  
3    "name": "Mike",  
4    "age": 30  
5  }
```

Key

: Colon separates key from value

, Comma separates elements



JSON - Nested Objects

```
1  // Nested Object Example
2  {
3      "name": "Mike",
4      "age": 30
5      "education": {
6          "college": true,
7          "mba": true,
8          "phd": false
9      }
10 }
```

JSON - Nested Objects

```
1  // Nested Object Example
2  {
3      "name": "Mike",
4      "storage": [
5          {
6              "id": 1,
7              "type": "backpack"
8          },
9          {
10             "id": 2,
11             "type": "suitcase"
12         }
13     ]
14 }
```



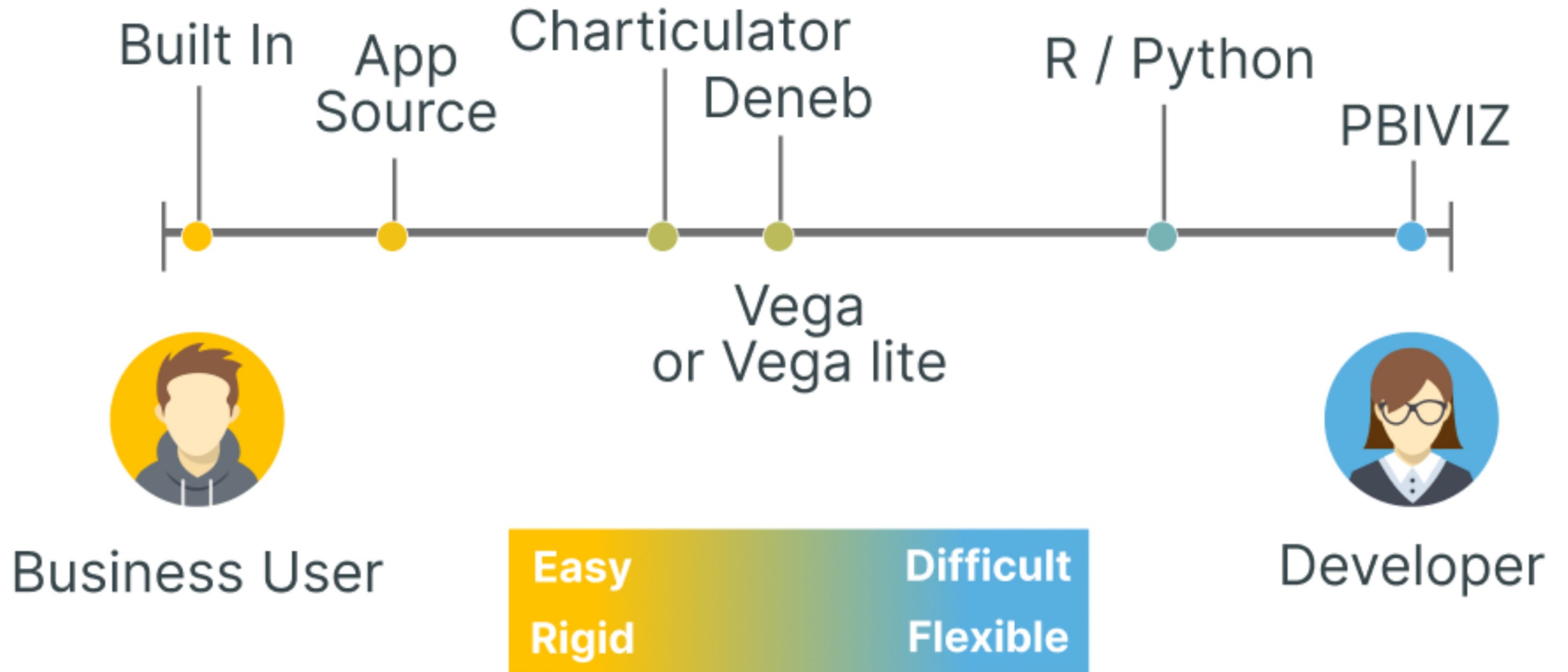
Introduction to



DENEb

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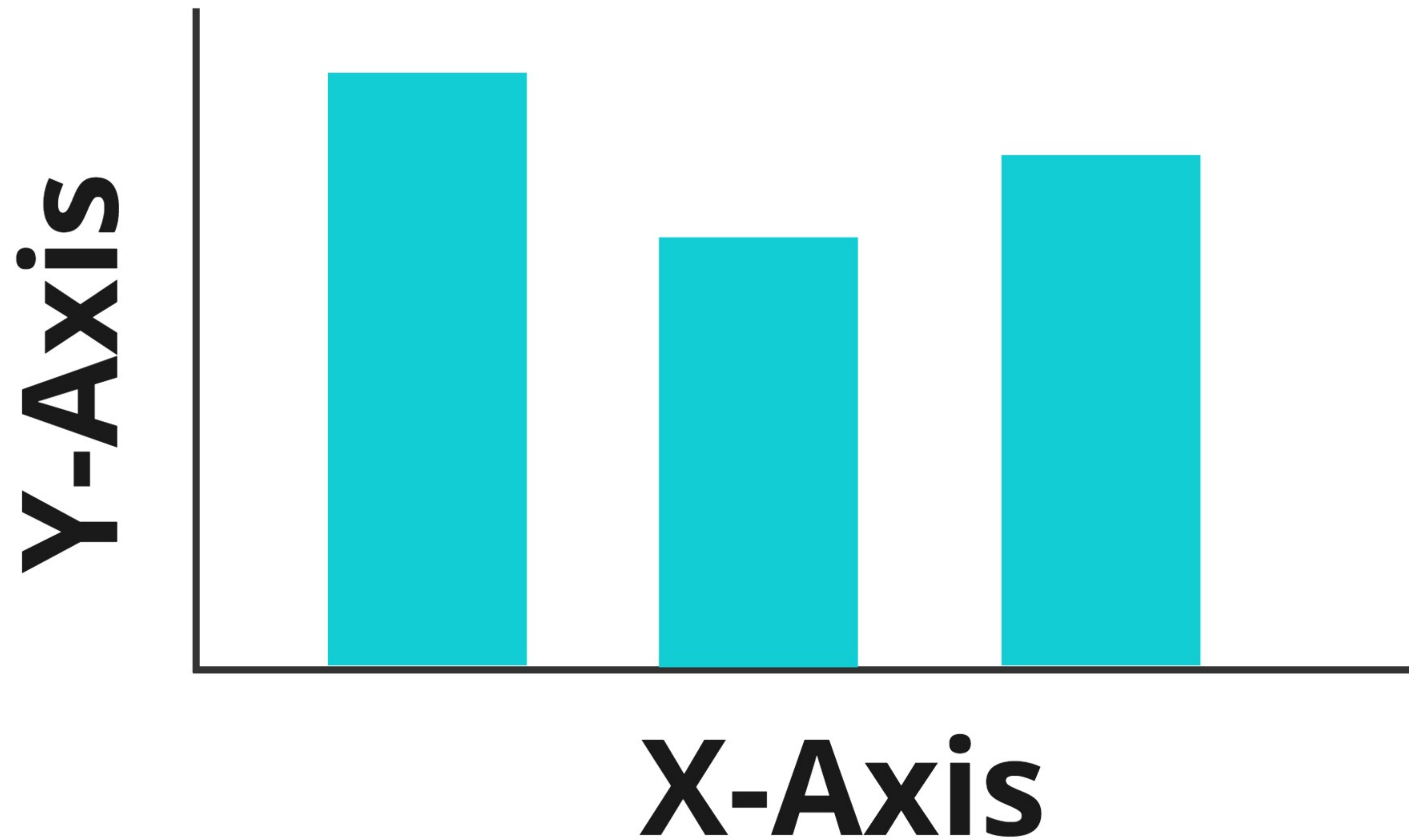
Visual Spectrum



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Shapes on a Page

Decomposition of a Visual



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Shapes on a Page

Tick Marks
Minor
Major

Y-Axis

Font Size
Font
Color
Labels
Label
Formatting



Fill Color
Fill Transparency

Hidden or Visible
Border Yes or No
Border Color
Border Thickness

Height

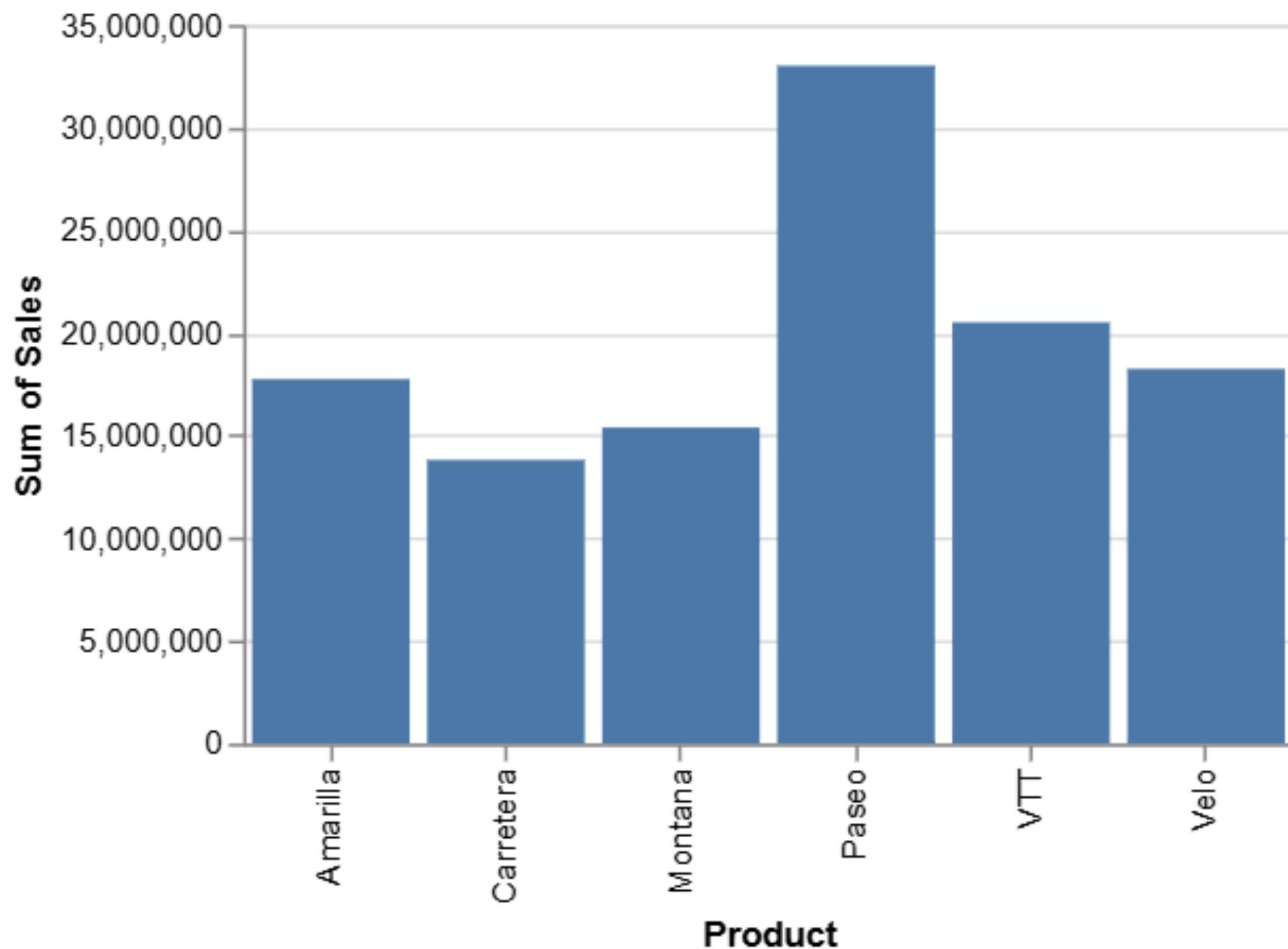
Width

Corner Rounding

X-Axis

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The Recipe



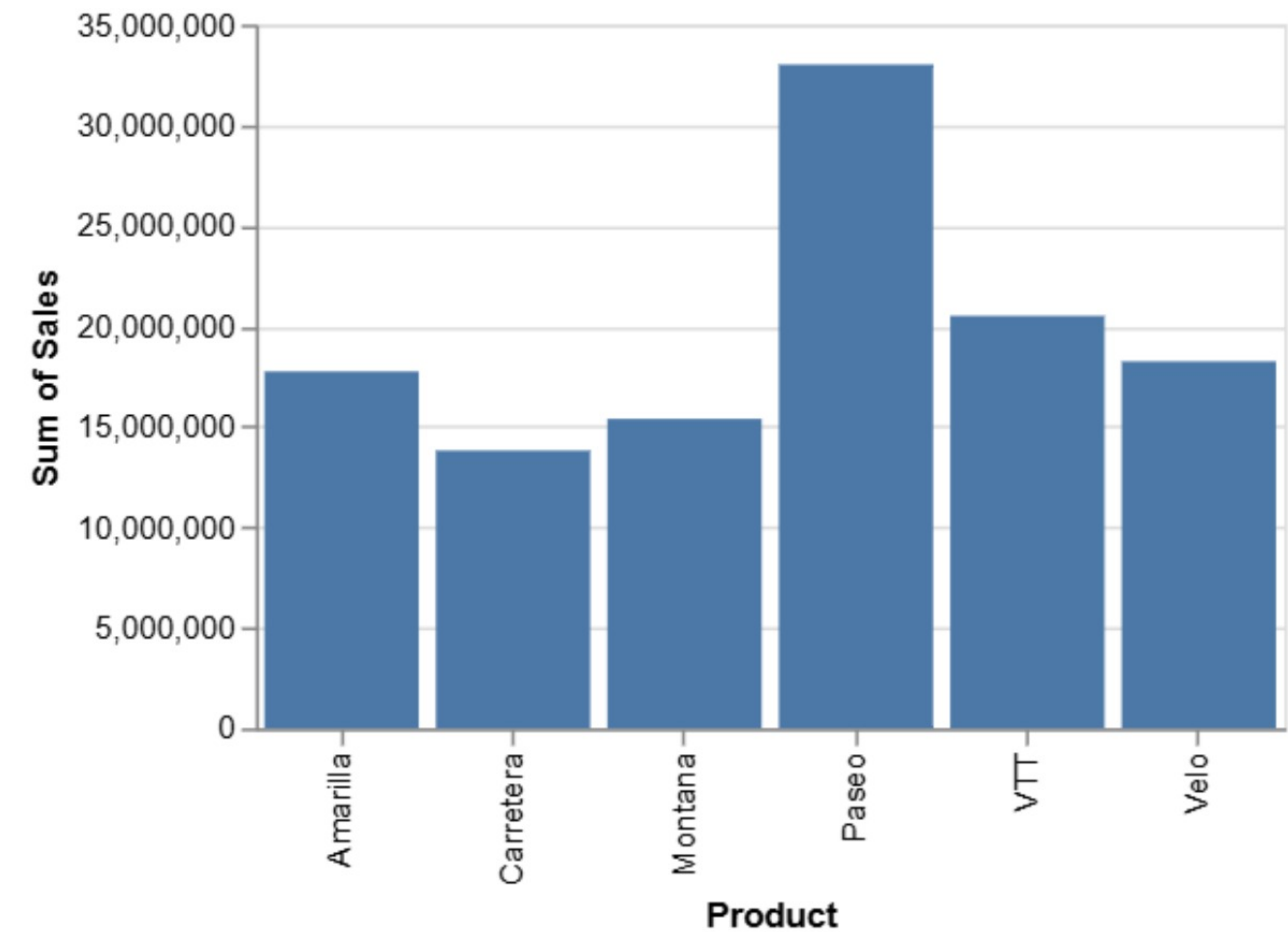
Product	Sum of Sales
Amarilla	17,747,116
Carretera	13,815,308
Montana	15,390,802
Paseo	33,011,144
Velo	18,250,059
VTT	20,511,921
Total	118,726,350

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The Recipe



```
1  {
2    "data": { "name": "dataset" },
3    "mark": { "type": "bar" },
4    "encoding": {
5      "x": {
6        "field": "Product",
7        "type": "nominal"
8      },
9      "y": {
10       "field": "Sum of Sales",
11       "type": "quantitative"
12     }
13   }
14 }
```



Product	Sum of Sales
Amarilla	17,747,116
Carretera	13,815,308
Montana	15,390,802
Paseo	33,011,144
Velo	18,250,059
VTT	20,511,921
Total	118,726,350

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Marking it Up



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Helpful Tips

Screen Resolution

Minimizing Power BI menus

Screen Snapping



Pain in the AI

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Ferry's Final Thought(s)

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BUDAPEST BI FORUM 2025

Workshop Evaluation



ws.budapestbi25.com